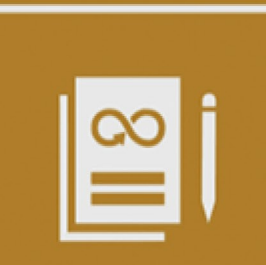
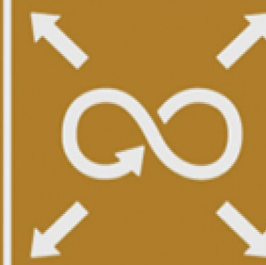




X

GOAL

12 RESPONSIBLE
CONSUMPTION
AND PRODUCTION





THE PERKS OF DATA-DRIVEN EVOLUTION

- **Operational Efficiency (Goal 12):** Precise data analysis cuts recycled material waste and optimizes logistics. Using Predictive Analytics, we accurately forecast demand, preventing overstock.
 - **Transparency and Trust (Veracity):** Data provides concrete proof of our environmental footprint. The Veracity of sustainability metrics is vital for customer trust in a green-image-driven market.
 - **Innovation and Personalization:** By analyzing data Variety (reviews, trends, purchases), we anticipate which green products will succeed, lowering risks and creating highly tailored solutions.
-

KEY METRICS TO MONITOR

TERRATEXTILES focuses on recycling and sustainability, so managing data Volume and Variety is vital across three main areas:

Area of Monitoring	Data Type (Example)	Analytical Function
Production & Sustainability	Energy/water use (IoT sensors), recycled material waste rate, supplier traceability.	Measuring progress toward UN Goal 12.
Inventory & Logistics	Warehouse stock duration, return rates due to defects or sizing, shipping costs.	Boosting efficiency and cutting logistic/environmental impact.
Customer & Engagement	Transactional data (sales), unstructure data (text/photo reviews, social feedback).	Understanding style preferences and satisfaction for artisanal garments.

REQUIRED PROFESSIONAL ROLES

To implement a Data-Driven strategy, the company must involve three key figures collaborating throughout the Data Journey:



- **Data Engineer (The Builder):**
 - **FOCUS:** Infrastructure and ETL.
 - **Role:** Builds the system to extract, clean, and load data into our Cloud Data Warehouse. Ensures proper management of the data Volume generated in-factory and via e-commerce.



- **Data Architect (The Schema Designer):**
 - **Focus:** Data Modeling
 - **Role:** Designs the database skeleton, defining the Logical and Physical Models. This ensures data is organized for analysis (tables, data types) and that consistency (keys) is maintained.



- **Data Analyst (The Interpreter & Storyteller):**
 - **Focus:** Descriptive Analysis and Visualization
 - **Role:** The bridge between data and business. Analyzes clean data, extracts insights, and translates them into clear reports and dashboards for the Board and operational teams.

THE DATA "JOURNEY"

Board Briefing: Data (whether an order or energy use) isn't analyzed directly; it follows a systematic 6-stage path:

- 1. Generate Data (Birth):** Data is created (e.g., a sale occurs or a sensor detects a waste spike).
- 2. Store Data (Storage):** Data is saved securely and scalably in our central Data Warehouse.
- 3. ETL (Cleaning):** The vital process where data is Extracted, Transformed (cleaned, standardized), and Loaded. This ensures data Veracity.
- 4. Aggregate Data:** Data is summarized (e.g., calculating the weekly average waste rate) for analysis.
- 5. Data Modeling:** Data is structured into linked "tables", ready to be queried.
- 6. Data Visualization (Communication):** The Data Analyst creates interactive dashboards.

For instance, a bar chart shows the waste rate per product line (for the Board), while a line graph tracks sales trends (for the sales team).

FUTURE SCENARIOS: INNOVATION WITH AI

Future integration of AI and Machine Learning (ML) elevates our sustainability and innovation:



- **AI for Circular Production:**

- Prescriptive Analysis can suggest to designers (based on recycled material inventory and predicted sales trends) which new garments are optimal to produce.
- **Benefit for Goal 12:** AI maximizes the use of available recycled material, reducing waste at the source (Responsible Production).



- **Automation for the Team:**

- AI models can automate simple data cleaning and standard report generation.
- This frees up the Data Analyst's time for complex and strategic analysis.

SUCCESS STORIES

Industry leaders prove that leveraging data is the ultimate key to sustainability:



Uses data analytics to optimize garment finishing processes, drastically cutting water consumption by shifting from intuition-driven to strictly number-driven operations.



Leverages predictive analytics and AI to forecast trends and adjust production. This minimizes aggressive markdowns and overproduction, mitigating the environmental impact linked to unsold inventory and return management.